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MD46

Current-Voltage Measurements of Au/Methylene Blue (MB)/n-Ge Schottky Contact in a Wide Temperature Range

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Abstract

Electrical properties of Au/Methylene Blue (MB)/n-Ge Schottky contacts were investigated in a wide temperature range from 125K to 400K. The device parameters such as barrier height, ideality factor and series resistance were determined using thermionic emission (TE) model and Cheung's method. The barrier height and ideality factor (n) values of the Schottky contact was determined from the current-voltage measurements and found to be 0.29 eV and 2.71 at 125K and 0.93 eV and 1.04 at 400K respectively. In the presence of inhomogeneity at the metal-semiconductor contact, the barrier height found to be decreased and ideality factor found to be increased with the decrease of temperature. From the Cheung's plot, the series resistance (R_s) was found to be decreased with increase of temperature. The barrier inhomogeneities has been elucidated by using the thermionic emission theory based on the assumption of gaussian distribution of Schottky barrier heights. However, the divergence in Schottky barrier heights of Au/Methylene blue/n-Ge Schottky contacts evaluated from current-voltage measurements indicates the deviation from the thermionic emission theory. The conventional Richardson plot between $\ln(I_0/T^2)$ vs. $1000/T$ gives an activation energy of 0.31 eV and Richardson constant (A^*) of $1.14 \times 10^{-9} \text{ A cm}^{-2} \text{ K}^{-2}$. The modified Richardson plot evaluated by assuming gaussian distribution of barrier shows an enhanced activation energy of 1.15 eV and Richardson constant (A^*) of $209.28 \text{ A cm}^{-2} \text{ K}^{-2}$ which is close to theoretical value of n-Ge. Current conduction mechanisms of Au/MB/n-Ge contact in a wide temperature range are resolved into four linear regions (Region-I to Region-IV) with different slope factors. This shows that the interfacial layer (MB) significantly influences the electrical properties of the Au/n-Ge contacts measured in a wide temperature range.

Keywords: Methylene blue, n-Ge, Schottky contact, Electrical properties, Current transport mechanisms, Barrier Inhomogeneity, Temperature Dependent

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